Algebra of Events and Conditional Probability

General Lecture Notes

These notes introduce the mathematical structures behind event algebra, conditional probability, independence, partitions, the law of total probability, and Bayes' formula. They are not solutions to a task list. Their purpose is to build the general language and reasoning tools needed before solving concrete problems.

Probability and Statistics for Introductory Students

Prepared as a self-contained LaTeX lecture note

Contents

1 Why Event Algebra Matters

Probability is not only about computing numbers. Before a probability can be computed, one must know precisely which event is being discussed. In elementary probability, events are sets of outcomes. Therefore, statements about events can be translated into set operations.

For example, a sentence such as "the user clicked the message and later made a purchase" is not merely ordinary language. It describes an intersection of two events. A sentence such as "the package was not delayed" describes a complement. A sentence such as "the device failed at least one of the two tests" describes a union.

Central idea

An event is a subset of the sample space. Logical words such as *and*, *or*, *not*, and *given that* correspond to mathematical operations on events.

The purpose of event algebra is to make this translation exact. Conditional probability then adds a second layer: it changes the reference population. Instead of asking how often an event happens among all outcomes, we ask how often it happens among outcomes that already satisfy another condition.

2 Sample Spaces and Events

2.1 Random experiments and elementary outcomes

A *random experiment* is a process whose result is not known in advance, but whose possible outcomes can be described. The set of all possible elementary outcomes is called the *sample space* and is usually denoted by Ω.

Sample space and elementary outcome

The sample space Ω is the set of all elementary outcomes of a random experiment. An elementary outcome $\omega \in \Omega$ is a single complete result of the experiment.

The meaning of an elementary outcome depends on how the experiment is recorded. If one records only whether a customer renewed a subscription, the outcome has two possibilities. If one records both the customer's channel and whether the subscription was renewed, the elementary outcome is a pair.

2.2 Events as subsets

An *event* is any subset of Ω. If $A \subseteq \Omega$, then the event A occurs when the realized outcome ω belongs to A.

$$A \text{ occurs} \iff \omega \in A.$$

Example: recording two binary properties

Suppose that a website records whether a visitor used a search bar and whether the visitor added an item to the cart. A possible sample space is

$$\Omega = \{(S, A), (S, A^c), (S^c, A), (S^c, A^c)\},$$

where S means "used search" and A means "added an item". The event S is not a single outcome; it is the set of all outcomes in which the visitor used search:

$$S = \{(S, A), (S, A^c)\}.$$

3 Basic Operations on Events

3.1 Complement

The complement of an event A, denoted by A^c, consists of all outcomes in Ω that do not belong to A:

$$A^c = \{\omega \in \Omega : \omega \notin A\}.$$

Complement rule

For every event A,

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A).$$

3.2 Intersection: "and"

The intersection $A \cap B$ is the event that both A and B occur:

$$A \cap B = \{\omega \in \Omega : \omega \in A \text{ and } \omega \in B\}.$$

3.3 Union: "or"

The union $A \cup B$ is the event that at least one of A and B occurs:

$$A \cup B = \{\omega \in \Omega : \omega \in A \text{ or } \omega \in B\}.$$

In mathematics, "or" is inclusive unless stated otherwise. Thus $A \cup B$ includes outcomes where both events occur.

3.4 Difference: "A but not B"

The difference $A \setminus B$ is the event that A occurs but B does not occur:

$$A \setminus B = A \cap B^c.$$

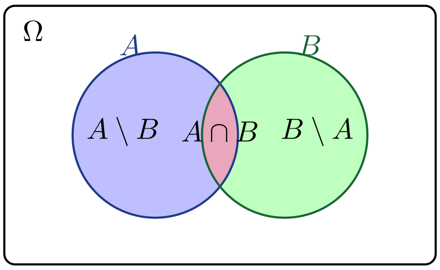

Figure 1: Two events divide the sample space into four basic regions.

4 The Four-Region Decomposition

Given two events A and B, the sample space can be decomposed into four disjoint regions:

$$A \cap B, \qquad A \cap B^c, \qquad A^c \cap B, \qquad A^c \cap B^c.$$

These four regions are disjoint and together cover the whole sample space.

Four-region decomposition

For any events A and B,

$$\Omega = (A \cap B) \cup (A \cap B^c) \cup (A^c \cap B) \cup (A^c \cap B^c),$$

and the four events on the right-hand side are pairwise disjoint.

This decomposition is the mathematical structure behind every two-way table.

4.1 Two-way tables

Suppose each observation is classified according to whether it satisfies property A and whether it satisfies property B. A general two-way table has the following form:

	B	B^c	Total
A	n_{11}	n_{10}	$n_{1\cdot}$
A^c	n_{01}	n_{00}	$n_{0\cdot}$
Total	$n_{\cdot 1}$	$n_{\cdot 0}$	n

The entries correspond to counts:

$$n_{11} = \#(A \cap B), \quad n_{10} = \#(A \cap B^c),$$
$$n_{01} = \#(A^c \cap B), \quad n_{00} = \#(A^c \cap B^c).$$

If the table is interpreted empirically, probabilities are obtained by dividing counts by the total number n:

$$\mathbb{P}(A \cap B) = \frac{n_{11}}{n}, \qquad \mathbb{P}(A \cap B^c) = \frac{n_{10}}{n}, \qquad \mathbb{P}(A^c \cap B) = \frac{n_{01}}{n}, \qquad \mathbb{P}(A^c \cap B^c) = \frac{n_{00}}{n}.$$

Reading a table conceptually

A two-way table should not be seen merely as a grid of numbers. It is a compressed representation of four disjoint regions of the sample space. Row totals, column totals, intersections, complements, and conditional probabilities are all obtained by selecting different parts of this decomposition.

5 Rules of Probability for Event Algebra

5.1 Addition for disjoint events

Two events A and B are disjoint, or mutually exclusive, if they cannot occur together:

$$A \cap B = \varnothing.$$

If two events are disjoint, then

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B).$$

Addition is not always direct

The formula $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$ is correct only when A and B are disjoint. If the events overlap, then the intersection is counted twice and must be subtracted once.

5.2 Inclusion-exclusion

For any two events,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

This is called the inclusion-exclusion formula.

Inclusion-exclusion for two events

The probability of "A or B" is the sum of the individual probabilities minus the overlap:

$$\boxed{\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).}$$

Example: app features

In a group of users, suppose

$$\mathbb{P}(A) = 0.55, \qquad \mathbb{P}(B) = 0.40, \qquad \mathbb{P}(A \cap B) = 0.25,$$

where A means "used feature A" and B means "used feature B". Then

$$\mathbb{P}(A \cup B) = 0.55 + 0.40 - 0.25 = 0.70.$$

Thus 70% of users used at least one of the two features. The value is not 95%, because users who used both features were counted twice in $0.55 + 0.40$.

5.3 Useful derived identities

The four-region decomposition gives several useful identities:

$$\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c),$$

$$\mathbb{P}(B) = \mathbb{P}(A \cap B) + \mathbb{P}(A^c \cap B),$$

$$\mathbb{P}(A \setminus B) = \mathbb{P}(A) - \mathbb{P}(A \cap B),$$

$$\mathbb{P}(A^c \cap B^c) = 1 - \mathbb{P}(A \cup B).$$

These identities are not separate tricks. They are all consequences of partitioning the sample space into disjoint regions.

6 Conditional Probability

6.1 Definition

Conditional probability measures the probability of one event under the assumption that another event has already occurred.

Conditional probability

If $\mathbb{P}(B) > 0$, the conditional probability of A given B is

$$\boxed{\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.}$$

The denominator $\mathbb{P}(B)$ is important. It means that after conditioning on B, the reference population is no longer the whole sample space Ω, but only the part of Ω inside B.

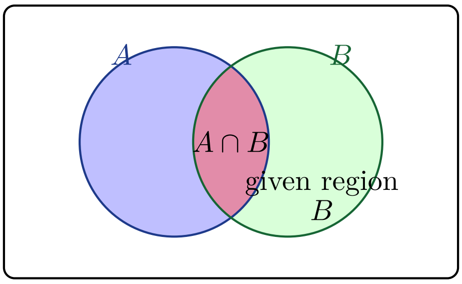

Figure 2: In $\mathbb{P}(A \mid B)$, the denominator is the whole event B, and the numerator is the part of B that also lies in A.

6.2 Conditional probability from counts

If a dataset contains n observations and B occurs in $n(B)$ observations, then

$$\mathbb{P}(A \mid B) \approx \frac{n(A \cap B)}{n(B)}.$$

In a two-way table, this means that conditioning on B restricts attention to the column corresponding to B. Conditioning on A restricts attention to the row corresponding to A.

Example: newsletter and purchase

A company observes 500 visitors. Let N mean "received a newsletter" and P mean "made a purchase". Suppose the table is:

	Purchase P	No purchase P^c	Total
Newsletter N	90	110	200
No newsletter N^c	60	240	300
Total	150	350	500

Then

$$\mathbb{P}(P \mid N) = \frac{90}{200} = 0.45,$$

because among newsletter recipients, 90 out of 200 purchased. However,

$$\mathbb{P}(N \mid P) = \frac{90}{150} = 0.60,$$

because among purchasers, 90 out of 150 had received the newsletter. These two fractions answer different questions.

6.3 Conditional probability is generally not symmetric

In general,

$$\mathbb{P}(A \mid B) \neq \mathbb{P}(B \mid A).$$

The two expressions have the same numerator $\mathbb{P}(A \cap B)$, but different denominators:

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \qquad \mathbb{P}(B \mid A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)}.$$

Common mistake

Do not switch the condition casually. The question "What is the probability that a selected person bought the product, given that they clicked the ad?" is not the same as "What is the probability that a selected person clicked the ad, given that they bought the product?"

7 Multiplication Rules

The definition of conditional probability can be rearranged. If $\mathbb{P}(B) > 0$, then

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B).$$

Similarly, if $\mathbb{P}(A) > 0$, then

$$\mathbb{P}(A \cap B) = \mathbb{P}(B \mid A)\mathbb{P}(A).$$

Multiplication rule

For events A and B,

$$\boxed{\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B \mid A) = \mathbb{P}(B)\mathbb{P}(A \mid B).}$$

For three events, one obtains a chain rule:

$$\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A)\mathbb{P}(B \mid A)\mathbb{P}(C \mid A \cap B),$$

assuming the relevant conditional probabilities are defined.

Example: two-stage quality control

Suppose a component first passes an automatic test with probability 0.92. Among components that pass the automatic test, the probability of passing a manual inspection is 0.97. Let A be "passes automatic test" and M be "passes manual inspection". Then

$$\mathbb{P}(A \cap M) = \mathbb{P}(A)\mathbb{P}(M \mid A) = 0.92 \cdot 0.97 = 0.8924.$$

Thus about 89.24% of components pass both stages.

8 Independence

8.1 Meaning of independence

Two events are independent if knowing that one of them occurred does not change the probability of the other.

Independence

Events A and B are independent if

$$\boxed{\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).}$$

If $\mathbb{P}(B) > 0$, this is equivalent to

$$\mathbb{P}(A \mid B) = \mathbb{P}(A).$$

If $\mathbb{P}(A) > 0$, it is also equivalent to

$$\mathbb{P}(B \mid A) = \mathbb{P}(B).$$

Independence is a probabilistic relation, not a visual or linguistic one. It says that the intersection has exactly the size one would expect if the two events had no probabilistic influence on each other.

8.2 Independence versus disjointness

Mutually exclusive events cannot happen together. Independent events can happen together, but the occurrence of one does not change the probability of the other.

Mutually exclusive does not mean independent

If A and B are disjoint and both have positive probability, then they are not independent. Indeed,

$$\mathbb{P}(A \cap B) = 0,$$

but

$$\mathbb{P}(A)\mathbb{P}(B) > 0.$$

Thus disjoint positive-probability events are dependent, not independent.

Example: independent events

A fair die is rolled and a fair coin is tossed. Let E be the event that the die result is even, and let H be the event that the coin shows heads. Then

$$\mathbb{P}(E) = \frac{1}{2}, \qquad \mathbb{P}(H) = \frac{1}{2}, \qquad \mathbb{P}(E \cap H) = \frac{1}{4}.$$

Since

$$\mathbb{P}(E \cap H) = \mathbb{P}(E)\mathbb{P}(H),$$

the events are independent.

9 Partitions of the Sample Space

9.1 Definition of a partition

A collection of events $B_1, B_2, \ldots, B_k$ is a partition of Ω if:

1. each outcome belongs to at least one of the events,
2. no outcome belongs to more than one of them,
3. each event has positive probability when used for conditioning.

Equivalently,

$$\Omega = B_1 \cup B_2 \cup \cdots \cup B_k,$$

and

$$B_i \cap B_j = \varnothing \quad \text{for } i \neq j.$$

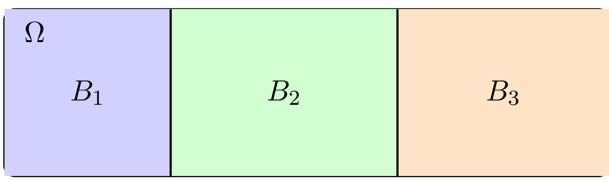

Figure 3: A partition splits the sample space into non-overlapping cases.

Partitions are useful because many problems naturally begin with a classification: channel of an order, machine used to produce a part, risk group of a customer, source of a message, weather condition, or category of a case.

10 Law of Total Probability

Suppose $B_1, B_2, \ldots, B_k$ form a partition of Ω. Any event A can be split across these cases:

$$A = (A \cap B_1) \cup (A \cap B_2) \cup \cdots \cup (A \cap B_k),$$

where the union is disjoint. Therefore,

$$\mathbb{P}(A) = \sum_{i=1}^{k} \mathbb{P}(A \cap B_i).$$

Using the multiplication rule,

$$\mathbb{P}(A \cap B_i) = \mathbb{P}(A \mid B_i)\mathbb{P}(B_i).$$

This gives the law of total probability.

Law of total probability

If $B_1, B_2, \ldots, B_k$ form a partition of Ω, then

$$\boxed{\mathbb{P}(A) = \sum_{i=1}^{k} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i).}$$

Example: delivery source and delay

An online store ships packages from three warehouses:

$$\mathbb{P}(W_1) = 0.50, \qquad \mathbb{P}(W_2) = 0.30, \qquad \mathbb{P}(W_3) = 0.20.$$

The delay probabilities are

$$\mathbb{P}(D \mid W_1) = 0.04, \qquad \mathbb{P}(D \mid W_2) = 0.07, \qquad \mathbb{P}(D \mid W_3) = 0.12.$$

The warehouses form a partition, so the overall delay probability is

$$\mathbb{P}(D) = 0.04 \cdot 0.50 + 0.07 \cdot 0.30 + 0.12 \cdot 0.20 = 0.065.$$

Thus the overall probability of delay is 6.5%.

Weighted average interpretation

The law of total probability says that an overall probability is a weighted average of conditional probabilities over all cases in a partition. The weights are the probabilities of the cases.

11 Bayes' Formula

Bayes' formula reverses the direction of conditioning. It allows us to compute $\mathbb{P}(B_j \mid A)$ from $\mathbb{P}(A \mid B_j)$, the prior probabilities $\mathbb{P}(B_j)$, and the total probability $\mathbb{P}(A)$.

Bayes' formula

If $B_1, B_2, \ldots, B_k$ form a partition of Ω and $\mathbb{P}(A) > 0$, then

$$\boxed{\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_{i=1}^{k} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}.}$$

For two events B and B^c, this becomes

$$\mathbb{P}(B \mid A) = \frac{\mathbb{P}(A \mid B)\mathbb{P}(B)}{\mathbb{P}(A \mid B)\mathbb{P}(B) + \mathbb{P}(A \mid B^c)\mathbb{P}(B^c)}.$$

11.1 Prior, likelihood, evidence, posterior

Bayes' formula is often interpreted using the following language:

Expression	Interpretation
$\mathbb{P}(B_j)$	prior probability of case B_j
$\mathbb{P}(A \mid B_j)$	likelihood of observing A under case B_j
$\mathbb{P}(A)$	evidence, or total probability of the observation
$\mathbb{P}(B_j \mid A)$	posterior probability after observing A

Example: server alert and rare critical error

A monitoring system checks server events. A critical error is rare:

$$\mathbb{P}(C) = 0.01.$$

If there is a critical error, the system sends an alert with probability

$$\mathbb{P}(A \mid C) = 0.95.$$

If there is no critical error, it may still send a false alert with probability

$$\mathbb{P}(A \mid C^c) = 0.04.$$

The total probability of an alert is

$$\mathbb{P}(A) = 0.95 \cdot 0.01 + 0.04 \cdot 0.99 = 0.0095 + 0.0396 = 0.0491.$$

By Bayes' formula,

$$\mathbb{P}(C \mid A) = \frac{0.95 \cdot 0.01}{0.0491} \approx 0.1935.$$

Thus, even after an alert, the probability of a critical error is about 19.35%, not 95%. The reason is that critical errors are rare, and false alerts can accumulate among the much larger group of non-critical cases.

11.2 The base-rate effect

The *base rate* is the prior probability of a case before observing additional evidence. Bayes' formula shows that a high detection probability does not by itself imply a high posterior probability. If the event is rare, false positives among the non-event cases may dominate.

Base-rate warning

A statement such as "the system detects 95% of true cases" describes $\mathbb{P}(A \mid C)$. It does not directly describe $\mathbb{P}(C \mid A)$. Confusing these two probabilities is one of the most common errors in interpreting diagnostic systems, alerts, and classifiers.

12 Tree Diagrams for Conditional Probability

A tree diagram is a convenient way to represent partitions and conditional probabilities. Each first-level branch represents one case in a partition. Each second-level branch represents an event conditional on that case.

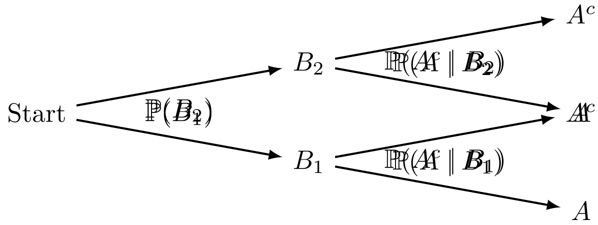

Figure 4: A conditional probability tree. Probabilities along a path are multiplied. Probabilities of disjoint paths leading to the same event are added.

The probability of one complete path is the product of probabilities along the path. For example,

$$\mathbb{P}(B_1 \cap A) = \mathbb{P}(B_1)\mathbb{P}(A \mid B_1).$$

If the event A can occur through several disjoint paths, we add the path probabilities:

$$\mathbb{P}(A) = \mathbb{P}(B_1)\mathbb{P}(A \mid B_1) + \mathbb{P}(B_2)\mathbb{P}(A \mid B_2) + \cdots .$$

13 How to Interpret Conditional Statements

Conditional probabilities should always be interpreted with the condition first. The phrase "given B" means that we are only looking inside B.

Expression	Correct interpretation
$\mathbb{P}(A \mid B)$	Among outcomes where B occurred, the proportion where A also occurred.
$\mathbb{P}(B \mid A)$	Among outcomes where A occurred, the proportion where B also occurred.
$\mathbb{P}(A \cap B)$	The probability that both events occur in the whole sample space.
$\mathbb{P}(A \cup B)$	The probability that at least one of the two events occurs.

Reading conditional probabilities in words

The expression $\mathbb{P}(A \mid B)$ should be read as: "within the group satisfying B, how common is A?" This wording prevents the most common confusion between $\mathbb{P}(A \mid B)$ and $\mathbb{P}(B \mid A)$.

14 A General Workflow for Problems

This section gives a general method, not a solution to any particular exercise.

14.1 When a two-way table is given

1. Identify the two events A and B.
2. Mark the four regions $A \cap B$, $A \cap B^c$, $A^c \cap B$, and $A^c \cap B^c$.
3. Divide counts by the total number of observations if probabilities are required.
4. Use row totals for probabilities conditioned on the row event.
5. Use column totals for probabilities conditioned on the column event.
6. Check independence by comparing $\mathbb{P}(A \cap B)$ with $\mathbb{P}(A)\mathbb{P}(B)$, or by comparing $\mathbb{P}(A \mid B)$ with $\mathbb{P}(A)$.

14.2 When a partition is given

1. Verify that the cases are disjoint and exhaustive.
2. Write the event of interest as a union across the cases.
3. Apply the law of total probability to compute an overall probability.
4. Apply Bayes' formula if the question asks for the probability of a case after observing the event.

14.3 When interpreting a result

A numerical probability is incomplete without a sentence explaining the reference group. For example, the statement

$$\mathbb{P}(A \mid B) = 0.72$$

should be interpreted as:

> Among observations satisfying B, 72% also satisfy A.

The condition determines the denominator. This is often more important than the numerical value itself.

15 Common Errors and How to Avoid Them

Error 1: Treating every "or" as direct addition

Use direct addition only for disjoint events. Otherwise apply inclusion-exclusion.

Error 2: Confusing conditional probabilities

$\mathbb{P}(A \mid B)$ and $\mathbb{P}(B \mid A)$ usually answer different questions. Always identify the denominator.

Error 3: Confusing disjointness with independence

Disjointness means the events cannot happen together. Independence means the occurrence of one event does not change the probability of the other. These are very different concepts.

Error 4: Ignoring base rates

A high value of $\mathbb{P}(A \mid B)$ does not necessarily imply a high value of $\mathbb{P}(B \mid A)$, especially when B is rare.

Error 5: Forgetting that a table represents a sample space

A two-way table is not just arithmetic. It encodes the four-region decomposition of the sample space. Every total and conditional probability must correspond to a clear region.

16 Compact Formula Summary

Concept	Formula
Complement	$\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$
Difference	$\mathbb{P}(A \setminus B) = \mathbb{P}(A) - \mathbb{P}(A \cap B)$
Inclusion-exclusion	$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$
Conditional probability	$\mathbb{P}(A \mid B) = \dfrac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$
Multiplication rule	$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B \mid A)$
Independence	$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$
Law of total probability	$\mathbb{P}(A) = \sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)$
Bayes' formula	$\mathbb{P}(B_j \mid A) = \dfrac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}$

17 Conceptual Checklist

Before computing, a student should be able to answer the following questions:

1. What is the sample space or reference population?
2. What events are being named?
3. Which operation is described by the words in the problem: and, or, not, but not, given?
4. Are the events disjoint, overlapping, or possibly independent?
5. If a conditional probability is involved, what is the denominator?
6. If there are several cases, do they form a partition?
7. Is the problem asking for $\mathbb{P}(A \mid B)$ or $\mathbb{P}(B \mid A)$?
8. If Bayes' formula is used, what are the prior, likelihood, evidence, and posterior?
9. Can the final number be interpreted clearly in words?

18 Closing Perspective

Event algebra and conditional probability are the bridge between elementary set language and probabilistic reasoning about data. Two-way tables, partitions, tree diagrams, and Bayes' formula are not separate techniques; they are different representations of the same structure.

The essential skill is not memorizing formulas in isolation, but recognizing which part of the sample space is being described. Once the event structure is clear, the formulas become natural consequences of set operations and probability rules.